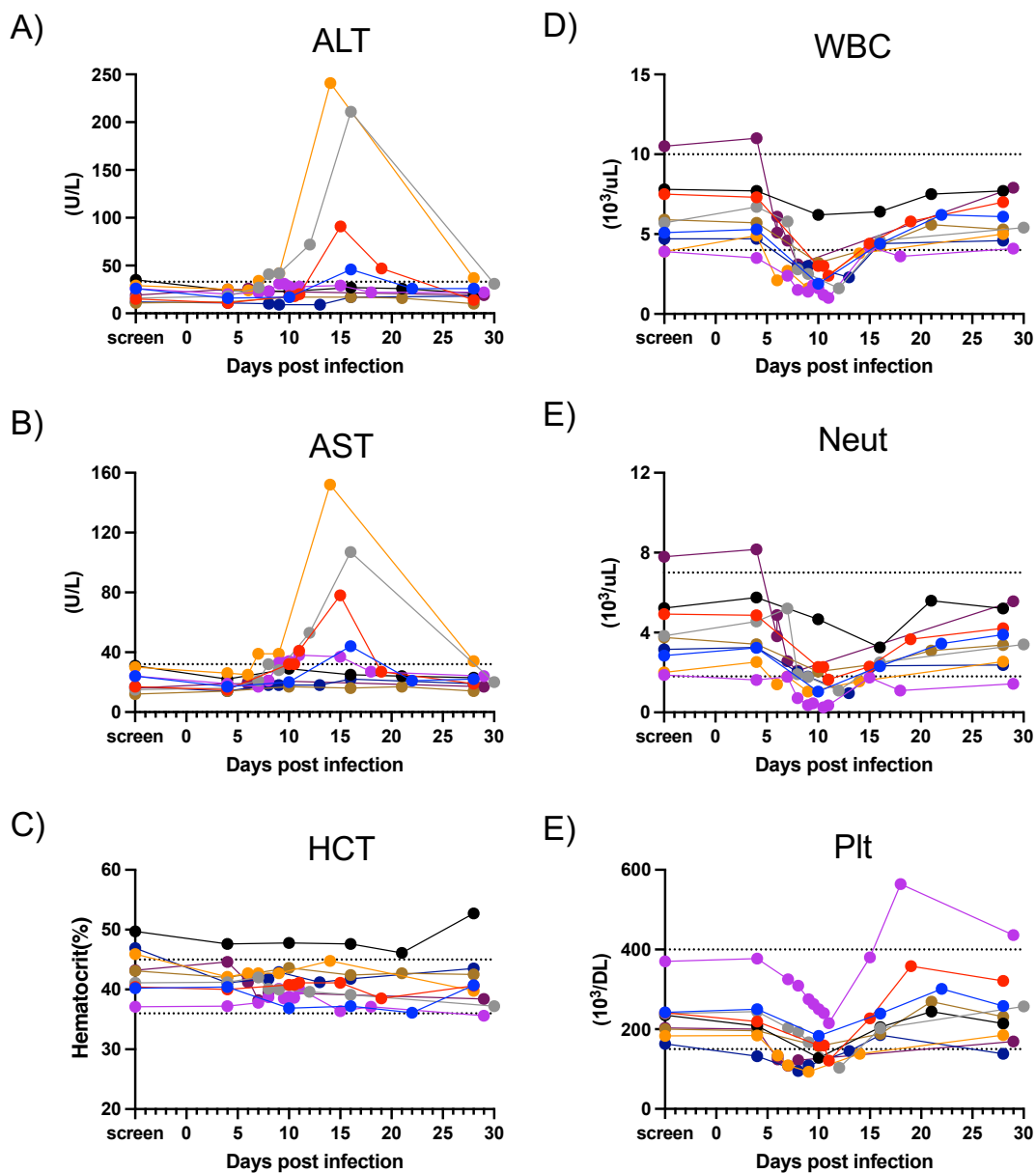
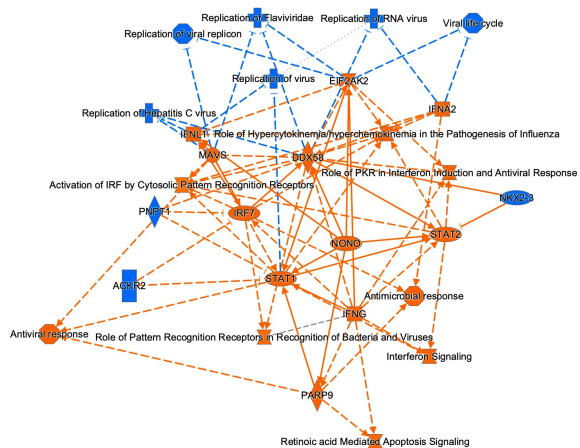

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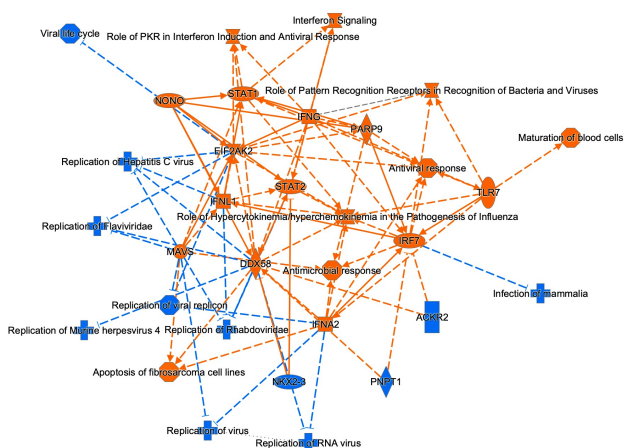


Supplementary Fig. 1. Timing and severity of clinically significant adverse events in response to CH53489 challenge. Raw lab values. Normal high and normal low for each lab shown in dotted lines

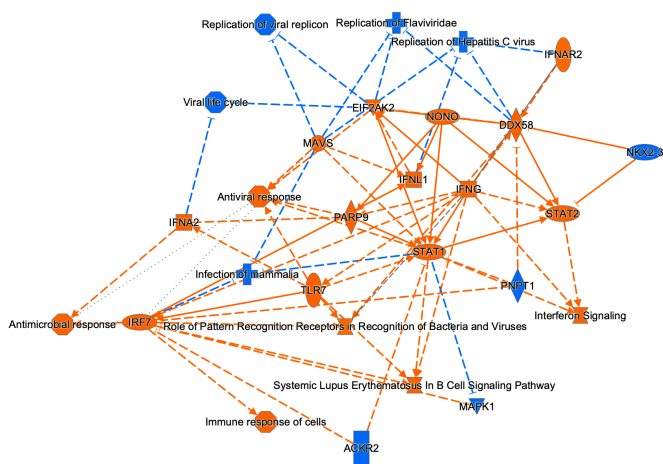
Day 6



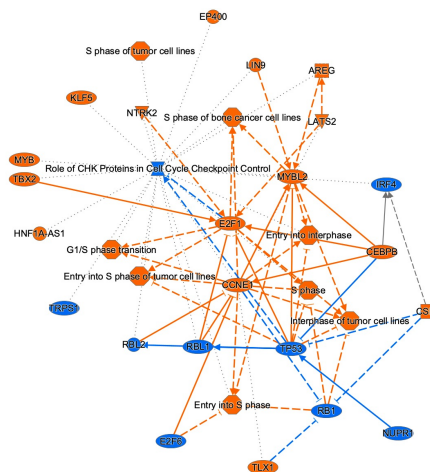
Day 8



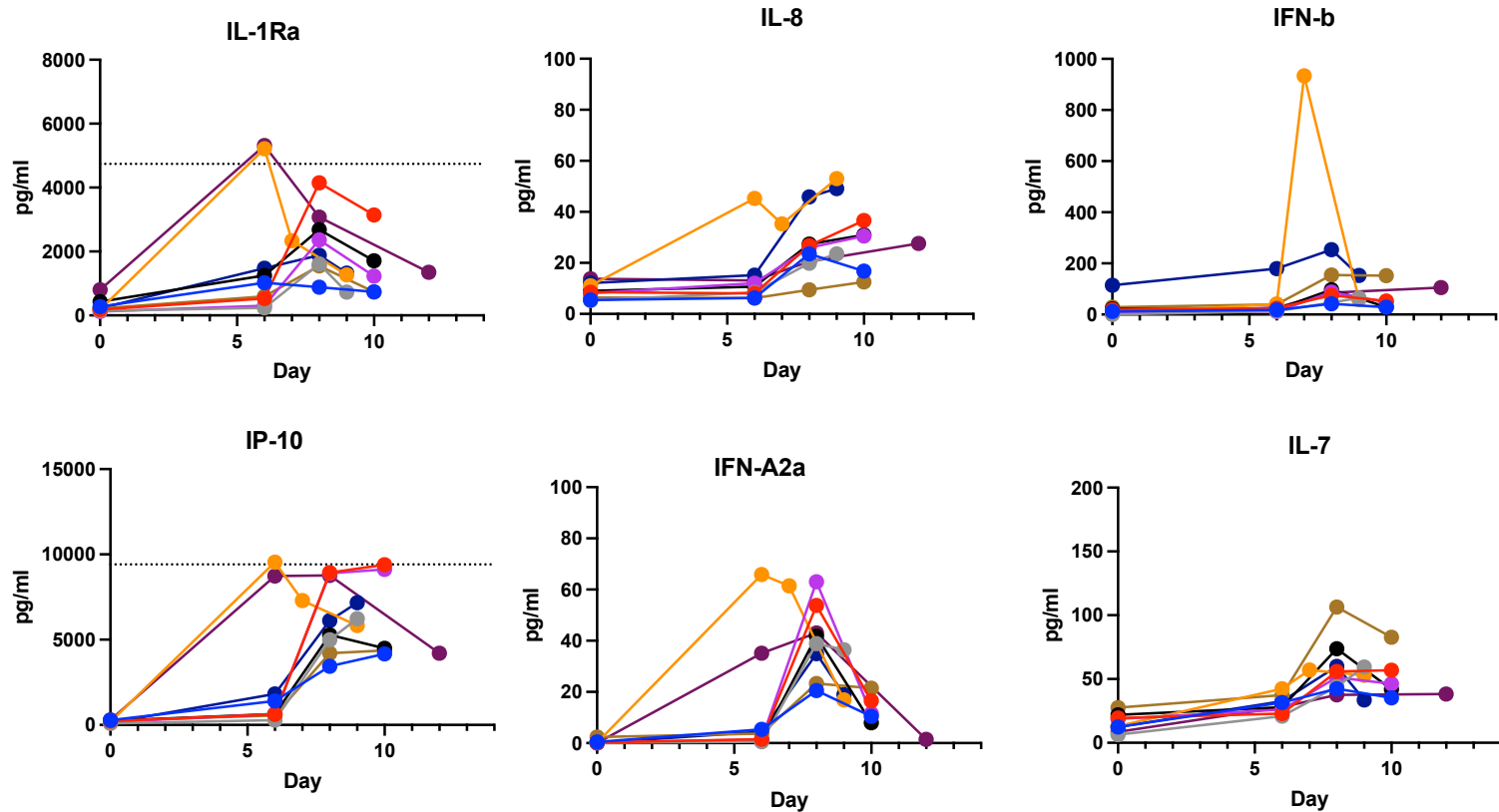
Day 10



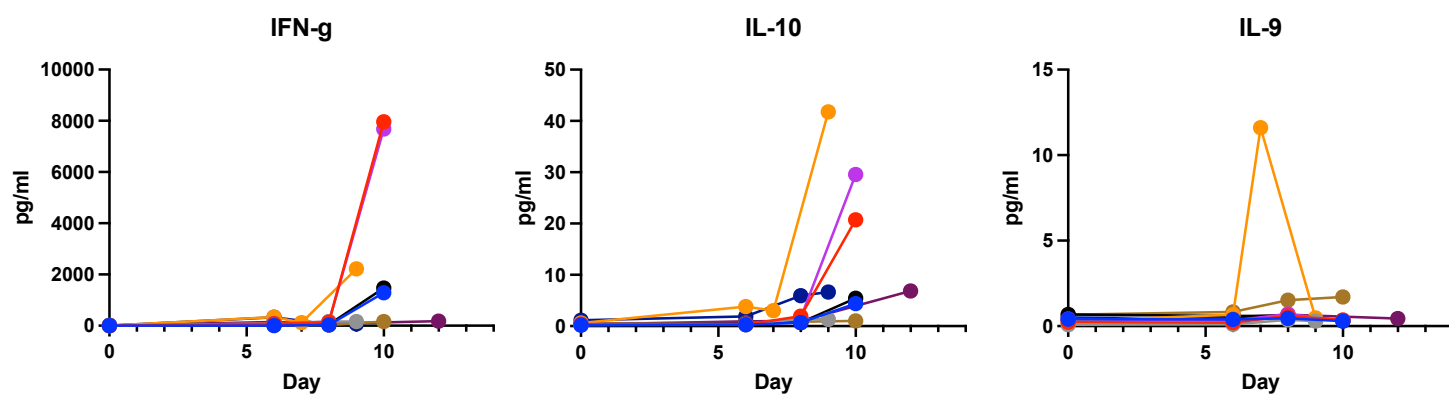
Day 12



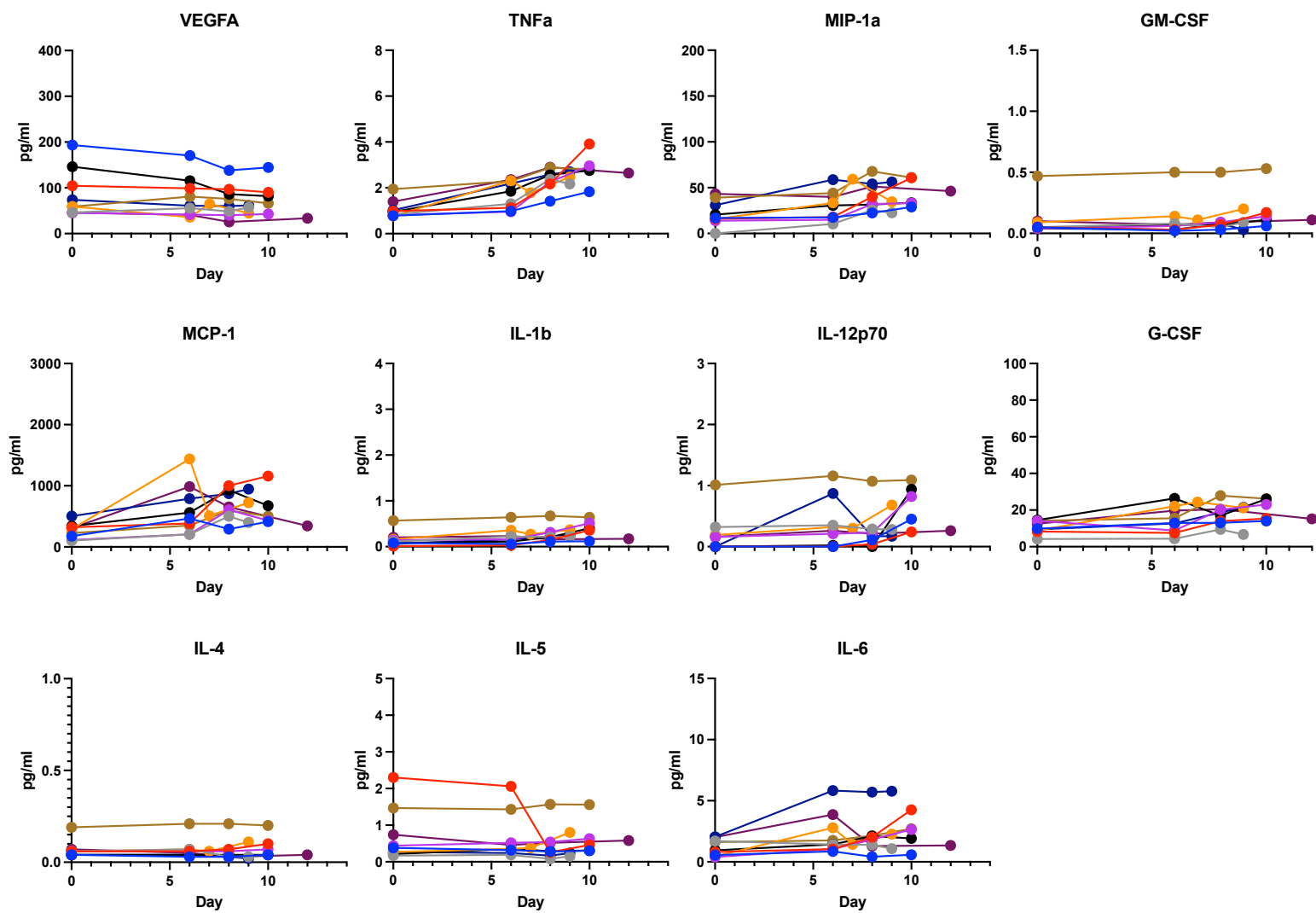
Supplementary Fig. 2. Graphical network summary of IPA terms identified across differentially expressed genes on days 6, 8, 10, and 14 post DENV-3 infection



Supplementary Fig. 3. Multiplexed cytokine analysis. Cytokines which exhibited induction days 6-8 post challenge. Dashed line indicates assay upper limit of quantification (ULOQ) for analytes with samples near or at the ULOQ.

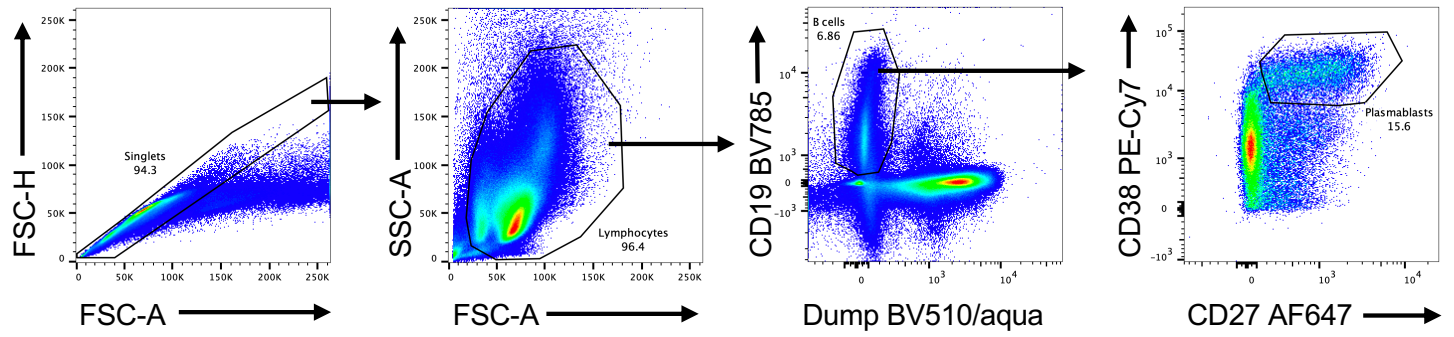


Supplementary Fig. 4. Multiplexed cytokine analysis. Cytokines which exhibited induction days 8+ post challenge

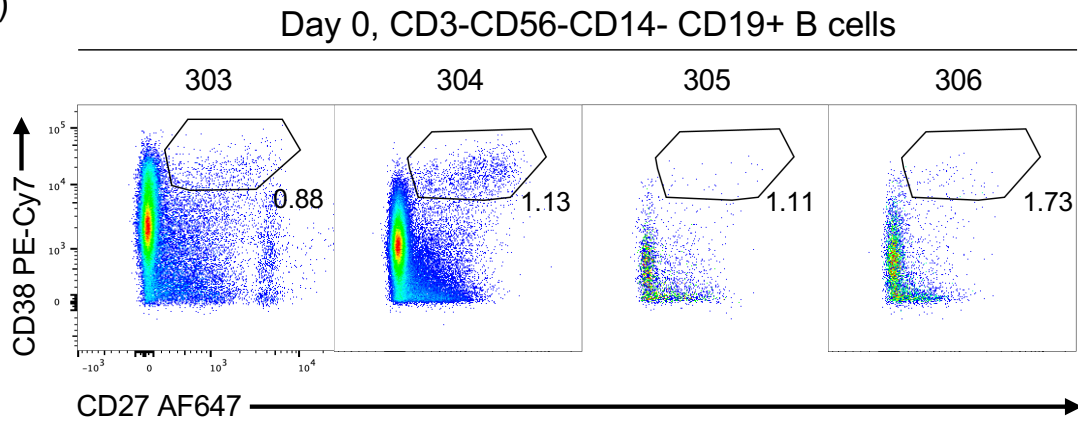


Supplementary Fig. 5. Multiplexed cytokine analysis. Cytokines which exhibited no change from baseline following challenge

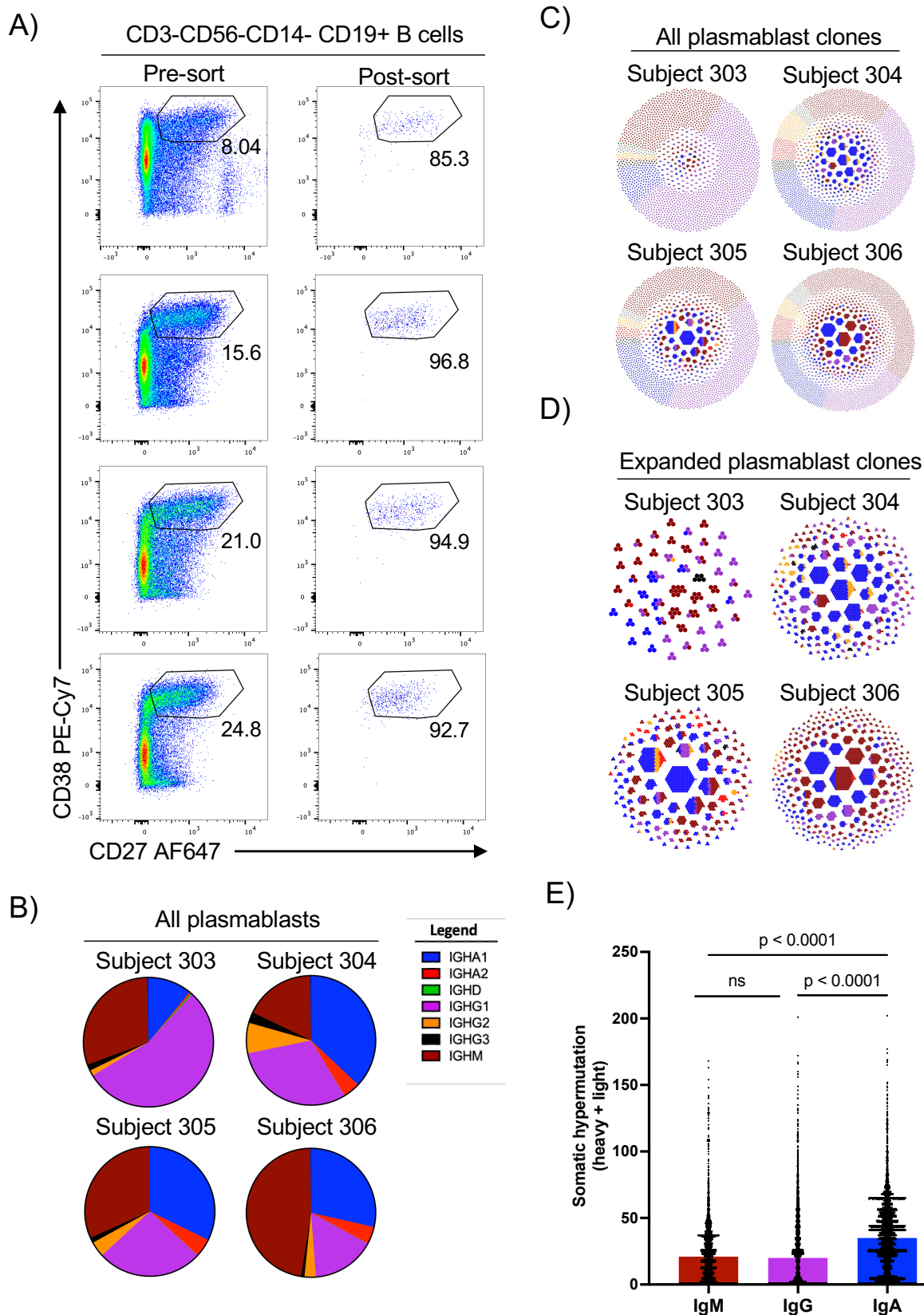
A)



B)



Supplementary Fig. 6. A) Gating scheme for flow cytometric isolation of DENV-3 elicited plasmablasts. **B)** Plasmablast frequency in all subjects on day 0.



Supplementary Fig. 7. Isolation and clonal characterization of DENV-3 elicited plasmablasts. **A)** Identification and post-sort purity of DENV-3 elicited plasmablasts **B)** Isotype distribution of DENV-3 elicited plasmablasts **C)** Clonal diversity and isotype distribution of all DENV-3 elicited plasmablasts **D)** Clonal diversity and isotype distribution of expanded ($n > 3$) DENV-3 elicited plasmablasts **E)** Somatic hypermutation burden of DENV-3 elicited plasmablasts separated by isotype. Bar indicates average (mean) SHM $\pm$ SEM. **** $p < 0.0001$ one-way ANOVA with correction for multiple comparisons. $n=3$ biologically independent samples

Supplemental Table 1. Study time and event table, days -90 to 7

Study Day	8	9	10	11	12	13	14	15	16	17-20^	21	22-27^	28	56	90	180
Visit window (days)	0	0	0	0	0	0	0	0	0	0	0	0	±2	±4	±7	±7
Phone Contact		X		X		X		X		X		X				
Physical exam, vital signs, temperature	X		X		X		X		X	X	X	X	X	X	X	X
Concomitant medications	X		X		X		X		X	X	X	X	X	X	X	X
Blood draw for safety labs			X						X		X		X			
COVID swab	X				X				X							
Blood draw for research assays	X		X		X		X		X	X	X	X	X		X	X
Saliva collection for research assay	X		X		X		X		X	X	X	X	X		X	X
AE assessment	X		X		X		X		X	X	X	X	X	X	X	X
SAE assessment	X		X		X		X		X	X	X	X	X	X	X	X
Illness Index card reviewed	X		X		X		X		X	X	X	X	X			
Illness Index card collected													X			

Supplemental Table 2. Study time and event table, days 8 to 180

[illegible]

Supplemental Table 3. Study time and event table in the event of hospitalization

Study Day	H1	H2-H8 ^b	Day 1 post discharge	Day 3 post discharge	Day 7 post discharge	Day 28	Day 56	Day 90	Day 180
Visit window (days)	0	0	0	±1	± 2	± 2	± 4	± 7	± 7
Physical exam, vital signs, temperature	X	X		X	X	X	X	X	X
Concomitant Medications	X	X		X	X	X	X	X	X
Blood draw for safety labs	X	X		X	X	X			
Blood draw for research assays	X	X		X	X	X		X	X
Phone contact			X						
Saliva swab	X	X		X	X	X		X	X
Illness Index card review	X	X		X	X	X			
Illness Index Card Collected					X	X			
AE assessment	X	X		X	X	X	X	X	X
SAE	X	X		X	X	X	X	X	X
Intake/output	X	X							

Supplemental Table 4. Sample selected for RNAseq analysis

Subject	Days analyzed	Peak RNAemia	Clinical outcome
301	0, 6, 8, 10, 14, 28	1.37E+05 (D6)	Non-hospitalized
302	0, 6, 8, 10, 15, 28	7.02E+08 (D8)	Hospitalized (D11)
303	0, 6, 8, 9, 16, 28	6.89E+07 (D8)	Hospitalized (D8)
304	0, 6, 8, 10, 15, 28	3.65E+07 (D7)	Hospitalized (D7)
305	0, 6, 7, 9, 14, 28	3.51E+08 (D6)	Hospitalized (D6)
306	0, 6, 8, 10, 14, 28	2.94E+05 (D6)	Non-hospitalized
307	0, 6, 8, 10, 14, 28	3.13E+04 (D8)	Non-hospitalized
308	0, 6, 8, 9, 13, 28	7.41E+05 (D6)	Hospitalized (D8)
309	0, 6, 8, 12, 14, 28	4.57E+07 (D6)	Hospitalized (D7)

Supplemental Table 5. Peptides used in IFN- γ ELISPOT assay

Virus	Antigen	Supplier	Catalog #
DENV-3: Sleman/1978	E (partial)	BEI	NR-511
DENV-3: Philippines/H87/1956	E (partial)	BEI	NR-9228
DENV-3: Philippines/H87/1956	NS1	BEI	NR-2753
DENV-3: Philippines/H87/1956	NS3	BEI	NR-2754
DENV-3: Philippines/H87/1956	NS5	BEI	NR-4204

Supplemental Table 6. Antibodies used for flow cytometry

Antibody	Clone	Manufacturer	Catalog #	Lot #	Dilution
CD38 PE-Cy7	HIT2	Biolegend	303516	B335356	1:50
CD27 AF647	O323	Biolegend	302812	B243264	1:200
CD56 BV510	5.1H11	Biolegend	362534	B340656	1:50
CD14 BV510	M5E2	Biolegend	301841	B332900	1:100
CD3 BV510	UCHT1	Biolegend	300449	B349804	1:50
CD19 BV785	HIB19	Biolegend	302239	B300756	1:200
Goat Anti-human IgA-AF647	Polyclonal	Southern Biotech	2050-31	G0919-V490B	1:1000
Goat Anti-human IgG-AF647	Polyclonal	Southern Biotech	2040-31	B3919-M950B	1:1000
Goat Anti-human IgM-AF647	Polyclonal	Southern Biotech	2020-31	D2219-R121	1:1000